OMAR SALAMA

Applied AI Engineer | LLM Products, RAG & AI Automation

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SUMMARY

AI Engineer focused on Agentic AI, LLM systems, and intelligent automation. Builds AI agents, RAG pipelines, multi-agent workflows, tool-calling systems, and business-process automations using LangGraph, n8n, structured outputs, and API-driven integrations. Experienced in connecting LLMs with CRM/ERP platforms, external tools, business logic, and enterprise data to build reliable end-to-end AI workflows. Strong foundation in machine learning, NLP, and computer vision, with hands-on experience taking AI systems from model development to automation, integration, and deployment. Focused on engineering practical agentic systems that automate complex workflows and enable AI-driven business processes.

EXPERIENCE

AI & Automation Engineer • Cultiv Bureau *Jul 2026 – Present | Cairo, Egypt*

- Designed **n8n automation workflows** that embed LLM intent classification and routing into CRM operations, as measured by faster lead handling, by integrating Bitrix24, Odoo, WhatsApp Cloud API, and REST/GraphQL webhooks.
- Built **RAG and document-intelligence pipelines** — chunking, embeddings, similarity search, and Google Cloud Vision OCR — to ground LLM responses in company knowledge for Arabic and English business documents.
- Cut response errors and token cost across production workflows by engineering structured prompts, **JSON Schema validation, and fallback logic**, then hardened delivery with API retry handling and Cloudflare Tunnel deployment.
- Worked directly with clients to translate operational requirements into automation and AI-enabled solutions, improving reliability across live workflow deployments.

NLP Research Assistant • Manipal Academy of Higher Education (MAHE) *Jul 2025 – Sep 2025 | Manipal, India*

- Researched a **multimodal AI credit-risk scoring system** combining structured financial data, emails, and financial documents.
- Applied **BERT/Transformer embeddings and feature fusion** for extracting and integrating financial risk signals.
- Explored **Explainable AI (SHAP/LIME)** and document understanding using **LayoutLMv3/Donut**.
- Co-authored an **NLP text-summarization research paper** under Prof. Ramakrishna.

AI & Machine Learning Trainee • Digital Egypt Pioneers Initiative (DEPI) *July 2025 – Jan 2026 | Cairo, Egypt*

- Completed intensive training in Python, Machine Learning, Deep Learning, Data Engineering, and AI deployment.
- Built end-to-end AI projects applying supervised/unsupervised learning, model development, evaluation, and deployment workflows.

PROJECTS

Bosla | AI Career Discovery and Growth Platform • GenAI Hackathon • MVP *Sep 2026 - Present*

- Co-developed a conversational career-discovery platform, building AI-personalization and full-stack features with React, TypeScript, and FastAPI, including provider fallback and user progress tracking.

TrueAgent — AI Employee & Multi-Agent Platform • Cultiv Bureau *Aug 2026 - Present*

- Designed a **multi-agent orchestration architecture** for CRM/ERP "AI employees," defining tool-calling, RAG, permissions, approval gates, and observability across Bitrix24, Odoo, and n8n integrations.

CohereVoice Studio — Speech AI & Meeting Intelligence • Repo *Aug – Sep 2026*

- Extended a speech-AI workflow routing multilingual audio through **ASR, diarization, and alignment** into bilingual subtitles and LLM-generated meeting minutes (OpenAI, Gemini, Groq).
- Shipped a Streamlit UI and a GPU-accelerated CLI batch pipeline with automated tests.

Shifaa — AI-Based Patient Monitoring Platform • Repo *Oct 2025 – Jul 2026*

- Engineered the hybrid real-time seizure-detection pipeline (**ViViT, Graph Neural Networks, OpenPose, ONNX Runtime, C++/Python**) for a privacy-first, on-premises hospital-monitoring graduation project, reaching **96.69% AUROC, 90.18% F1, and 37+ FPS**.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence (Dual Degree) — Ain Shams University & University of East London
Sep 2022 – Jul 2026 | Cairo, Egypt | GPA 3.3/4.0

IGCSE — Rowad College
Sep 2019 – Jun 2022 | Cairo, Egypt

TECHNICAL SKILLS

Agentic AI & LLM: RAG, Multi-Agent Orchestration, LangGraph, Tool Calling, Prompt Engineering, MCP

AI/ML, CV & NLP: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, Vision Transformers

Automation & Integration: n8n, Bitrix24, Odoo, REST/GraphQL APIs, Webhooks, JSON Schema

Programming & Infra: Python, C++, TypeScript, PostgreSQL, Docker, AWS, Git

Certifications & Achievements: AWS Certified Machine Learning – Specialty • HCIA-AI V3.5, Top 5 Finalist – ARABCAB 2026